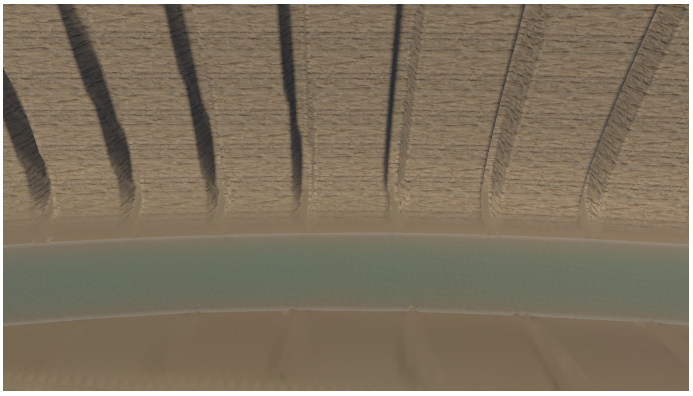
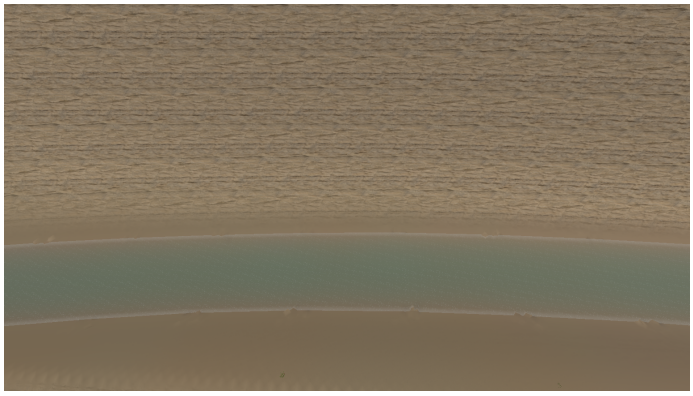
Terrain that stays joined

Repairing broken cliffs, missing ground and visible chunk boundaries.



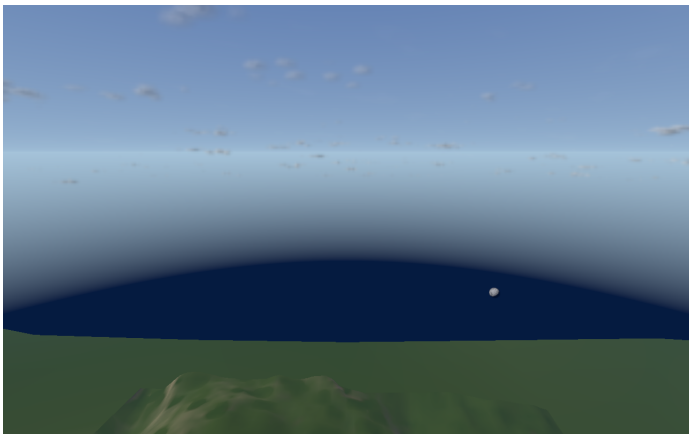
Matched cliff test, before: repeated artificial comb-like ridges.



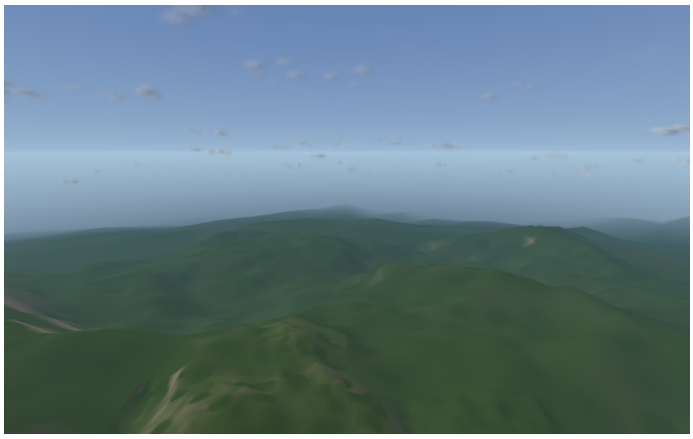
Matched cliff test, after: the artificial projections removed.

Solid ground while detail arrives

Loading prioritizes coverage before fine detail, then refines nearby ground. More accurate fallback heights retain hills instead of a plain surrounding apron. Smooth updates gradually change eligible distant slopes; interactive ground and riverbanks change immediately to preserve contact.



Matched forward flight, before: terrain cuts off ahead of the player. This is a separate location from the cliff test.

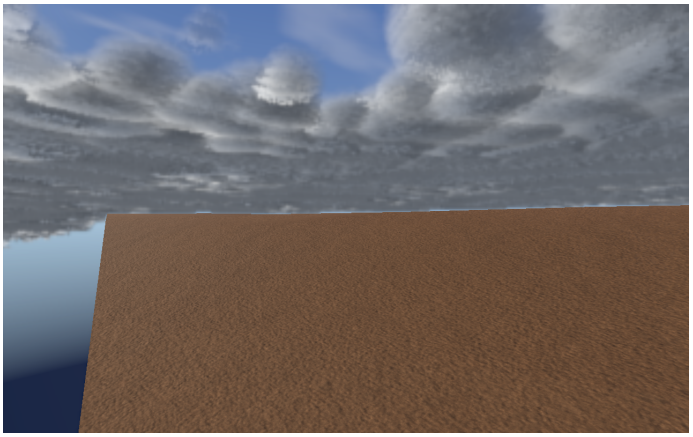


After: continuous ground and nearby relief on the same flight route, with hard chunk updates enabled.

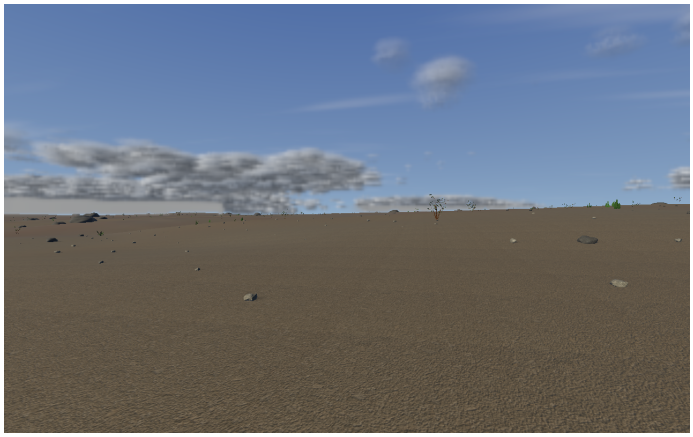
The 260-second flight at 600 m/s crossed a terrain chart without missing resident ground below or ahead of the player. A separate Balanced run also kept coverage while finer ground arrived. Coarse distant terrain remains approximate.

A complete first view

Actual menu-to-world entry; random starting points differ.



Before: gameplay opened with one ground tile beside a blue void.

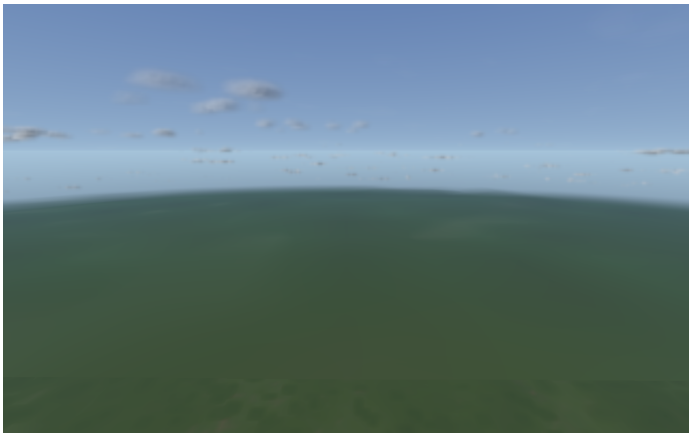


After: the first playable view has continuous surrounding ground.

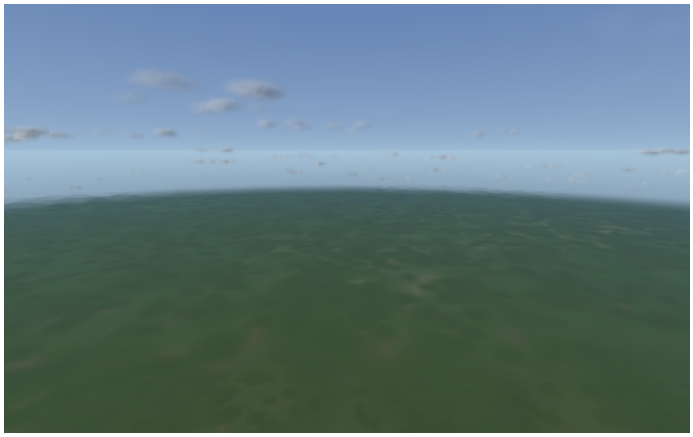
Load the surroundings before taking control

The loading screen now stays up until usable ground and coarse coverage of the visible area are ready. Controls, survival and world time stay paused, then resume together. Fine terrain and vegetation continue streaming.

The shared tree texture bank and all eight leaf shapes are prepared during loading. This moves shared cold setup away from the first visible tree; model creation and storage growth remain performance risks.



Balanced loading: the old background stayed mostly plain.

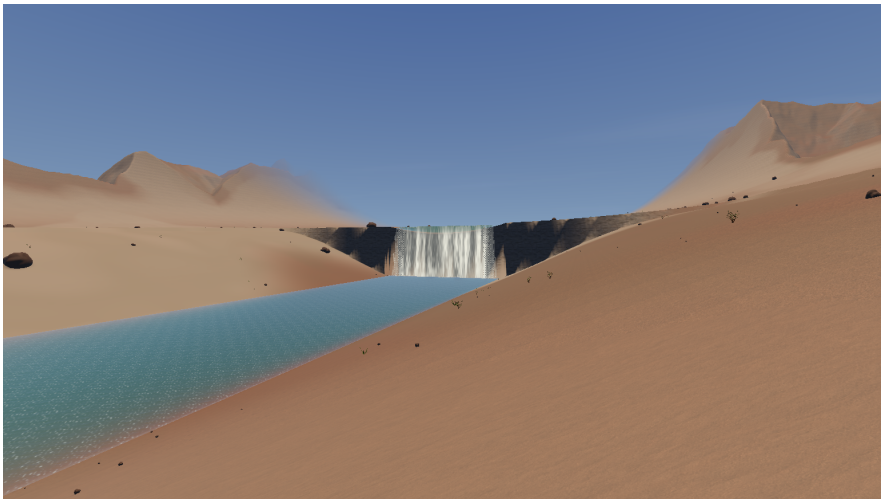


Balanced loading: nearby fallback relief now arrives during flight.

The same short flight now has zero missing forward-ground probes. This is a loading and coverage result, not an FPS benchmark.

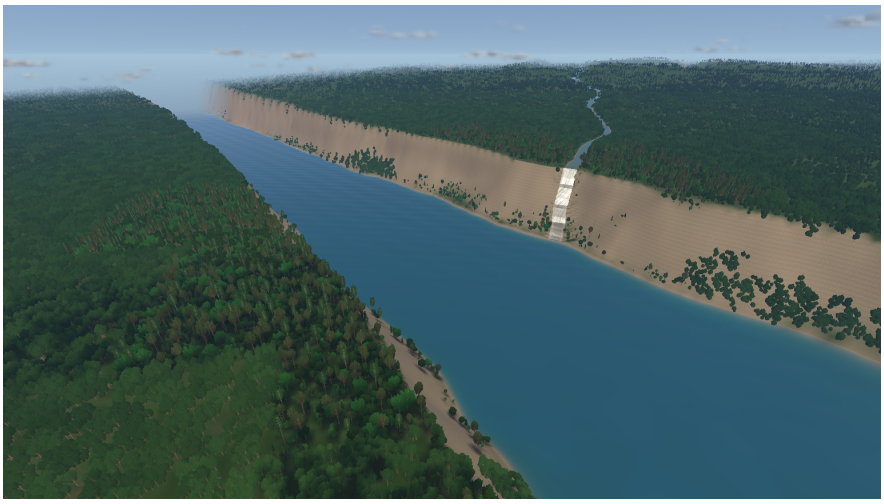
Water, woodland and contact

Separate generated locations; the original reported world was unsaved.



Latest waterfall review: paired bank spikes removed; simple channel and fall shapes remain visible.

Ground support follows the terrain triangles actually on screen. Actual player collision checks in both canyon and waterfall settings stayed above steep ground when an adjacent obstacle redirected movement. Riverbeds remain below the water in the checked channel interiors and boundaries.



Fjord forest review, separate from the canyon and river flight: distant trees retain physical scale and upright silhouettes.

Trees anchor to the displayed ground as terrain changes. Distant forest colour reflects crown coverage; oversized tree cards were replaced. Tree placement is spread across frames without exposing incomplete groups, and completed placement matches the original trees exactly.

Still visible: some aerial forests look too sparse, leaf edges can look thin, and ground plants repeat. Canyon ledges also remain too evenly spaced. These are remaining appearance limits; the repair does not claim finished landscape art.

Performance and limits

Representative checked routes; performance remains unfinished.

Travel and turning

Route / checked build	Average FPS	p99 frame	Worst frame
Dry flight / original	54.9	35.9 ms	53.0 ms
Dry flight / repaired	127.1	23.7 ms	42.0 ms
Jungle / checked build 3d	165.3	14.7 ms	23.6 ms
Temperate forest	172.5	13.9 ms	23.5 ms
Boreal forest	192.7	12.2 ms	20.3 ms
Savanna	223.1	9.1 ms	27.8 ms

1080p, Medium vegetation, 6 km view; Chrome video shared the PC. Dry grassland flight: 200 m/s at 500 m altitude. Forest/savanna: 8 m/s travel with rapid turns. These are representative routes, not the worst place on the planet. p99 means 99% of frames were faster than that time. A strict 100 FPS minimum requires every frame below 10 ms: this remains unmet.

An earlier jungle run reached an 80.6 ms frame, including 69.3 ms placing forest data. It did not recur in the latest sample but has not been proven fixed. A 14.5 ms buffer growth occurred during warm-up only; no slow growth was logged in the measured interval. The latest maximum is not a guarantee, and no average-speed gain is claimed.

Loading settings: 6 km view

Empty spectator load	Maximum (hard)	Balanced
Preparation before streaming	14.1 s	14.7 s
Nearby terrain, after preparation	1.2 s	1.6 s
600 m terrain + forest, after prep	11.1 s	15.7 s
Entire view, after preparation	32.8 s	39.4 s

Preparation is additional to the streaming milestones. These empty-spectator checks exclude NPCs and camps; they are not full-game startup timings. Screenshot checks and 60 FPS pacing measure loading and continuity, not gameplay FPS. Startup still has a noticeable first-time asset stall.

Playtest checkpoint and remaining work

The normal shortcut selects the checked Mono build a365. Actual entry, long hard-update travel, shorter Balanced travel, cliff contact and waterfall views were checked. Remaining work includes intermittent CPU stalls, coarse distant detail and repeated canyon ledges. The strict 100 FPS minimum remains unmet.